

User Experience Design and Research in Games

Lennart E. Nacke
HCI Games Group, Stratford School
of Interaction Design and Business,
University of Waterloo
Waterloo, Ontario, Canada
lennart.nacke@acm.org

Pejman Mirza-Babaei
Bluemoon UX Research and
Development and Ontario Tech
University
Ontario, Canada
pejman@uoit.ca

Anders Drachen
SDU Metaverse Lab, Maersk
McKinney Moeller Institute,
University of Southern Denmark
Odense, Denmark
adrac@mmmi.sdu.dk

ABSTRACT

This online course trains how user experience (UX) methods are used in a game context. The course consists of three units: UX design for games, games user research, and game analytics. The course material comes from the "Games User Research" book published by Oxford University Press, and the book's editors will teach it. This course focuses on UX design and research for game development. Students will learn the skills they need to recognize, analyze, and understand player feedback so they can make valid decisions about how to design games. Through exercises and assignments, participants will learn how to identify factors that affect how a player plays a game and how to incorporate feedback into their design process. Participants will learn ways to get information from players, such as through direct observation, interviews, and surveys. Participants will be equipped with skills, knowledge, and tools to understand players to create engaging games.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods; User centered design; • **Applied computing** → Computer games.

KEYWORDS

Games User Research, UX Design, Game Design, Game Analytics, UX for Games

ACM Reference Format:

Lennart E. Nacke, Pejman Mirza-Babaei, and Anders Drachen. 2023. User Experience Design and Research in Games. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3544549.3574181>

1 BENEFITS

Students will learn how to design UX and conduct UX research for games. Specifically, we aim for participants to take away the following skills from this course:

- ✓ Understanding UX design approaches to video games.
- ✓ Learning how to conduct games user research. An overview of the most common approaches

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3574181>

- ✓ Getting detailed insights into game analytics
- ✓ Identifying & addressing user pain points in game design.
- ✓ Gaining holistic understanding of UX design process for games from research to implementation.
- ✓ Communicating design recommendations and justifying decisions based on user data.
- ✓ Incorporating user feedback and data analysis into game design.
- ✓ Conducting user testing and research for games, including method selection, analysis, and interpretation.

This course will provide an introduction to UX design and research for video games. People who attend the CHI conference can learn much about how to use UX design techniques in their own game development processes. They can also get an introduction to different methods, such as user testing, usability analysis, and making decisions based on data. By taking this course, people will learn more about current trends and the best ways to design and research UX for video games. They will learn what leads to an improved player experience. Attendees may also gain skills that are in high demand by employers looking for experienced professionals with expertise in this field.

2 INTENDED AUDIENCE

This course is ideal for people who are just starting out in their UX design and research careers, as well as students who are interested in the field. It will provide a comprehensive overview of topics related to user experience design and research for video games, including how to identify user requirements, create effective designs based on player feedback, conduct usability testing, and analyze data from play tests. The course also talks about things like game analytics, prototyping tools, and other ways to make games better. Participants will learn a lot about how the games industry works, which will help them make smart decisions about UX design and research when working on their own projects or with others. This course is designed to be accessible to anyone, regardless of their prior experience. To take part, you don't need to know anything about the subject or have any special technical skills. All that is needed is an interest in learning about user experience design and research in games.

- Junior researchers wanting to learn more about UX design and research in video games.
- Graduate students wanting an overview of what's possible in UX research and design for games.
- Quantitative researchers wanting to branch out into game analytics.

- Designers and programmers of video games that wish to enhance player experience.
- Marketing professionals looking to understand player behaviour and preferences to target better and retain players.
- Product managers who wish to better comprehend user wants and preferences in the gaming business.
- Researchers in fields such as psychology or human-computer interaction who want to learn more about UX research in the context of games.
- Professionals in other fields who are interested in applying UX design and research principles to games or other interactive products.

3 PREREQUISITES

The course is a follow-up on an early courses taught by the same instructors [2, 3, 6] and based on their published book [1]. Having read the book is not a pre-requisite but will provide a nice knowledge angle towards the course. There are no prerequisites for this course. The participants are expected to have a computer with a stable Internet connection, online video software with microphone and webcam available. It is also recommended that they have accounts and logins created for Notion and Miro before the beginning of the course.

4 CONTENT

The course is structured into three online units (see Table 1, each 75 minutes long), the first one UX design for video games, the second one on an overview of games user research methods, the third one on game analytics.

Table 1: Structure of three online course units.

Unit 1	UX Design for Games
25 mins	Introduction, Goals, and Icebreaker activity
50 mins	Discussion of UX Design in Games [4, 10] (Nacke and Mirza-Babaei)
Unit 2	Games User Research Methods
15 mins	Discussion: Which methods for what research questions? [2] (Everyone)
30 mins	Methods on a budget and working with indies [5, 7, 8] (Mirza-Babaei)
30 mins	Lab-based user research (Nacke)
Unit 3	Game Analytics
15 mins	Recap and outlook [1] (Drachen and Mirza-Babaei)
60 mins	Guided discussion and presentation on game analytics (Drachen)

4.1 Unit 1: UX Design for Games

The main goal of this unit is to look at different ways to design games from the user's point of view. We will talk about how important it is to design games in an iterative way and take into account different types of players. The primary focus will be on creating a positive user experience, taking into account each player's individual perspectives and needs [11]. Additionally, we will examine current trends in UX research for games and how it can be used to inform design decisions. Finally, we will review best practices for playtesting game features prior to launch to ensure that the finished product meets all criteria set forth by the team during development.

4.2 Unit 2: Games User Research Methods

The instructors will talk in depth about the user research methods in their book [1]. This book gives games user researchers helpful information about many methods they are already using, such as usability testing, interviews, heuristic evaluations, and surveys. By learning more about these methods and using them in their own projects, students can learn more about how to make games that players will enjoy the most. We will emphasize two approaches in particular. The first is a quick and iterative way for independent game developers to test mobile and casual games. This means testing raw prototypes of ideas and using testers' comments to improve the game before putting out a final version. The second approach is lab-based traditional user research experimentation, which typically consists of conducting longer research studies with participants to identify issues related to the design, layout, or mechanics of a game. By using both methods, developers can learn more about the preferences of their target audience and make sure that the overall user experience is good enough before the product is released.

4.3 Unit 3: Game Analytics

In the last section, we'll talk about how quantitative methods can be used to measure UX in games. This includes measuring behavioural telemetry[9], which is data that records how users interact with a game. The goal of these metrics is to gain insights into the behaviour and preferences of players so that developers can improve the overall UX design. For example, if developers know how engaged players are in certain areas and how they play, they can figure out where they need to make changes or add features to improve gameplay and give players the best player experience possible. Also, looking at data from game analytics can help figure out which parts are most crucial for creating interesting experiences. Behavioural measures such as time and money spent on a game can provide valuable insights into how players interact with a game. These analyses may assist a game production studio in figuring out what aspects of a game players find interesting, where they could become frustrated or perplexed, or what kinds of experiences encourage them to buy things or keep playing. By understanding these behavioural patterns, user researchers can increase player satisfaction and company revenue. Quantitative methods such as behavioural telemetry in conjunction with qualitative research techniques give games user researchers powerful tools.

5 PRACTICAL WORK

The coursework in the three units will feature components of practical work, such as online discussions about games user research and working with Notion and Miro. Our website at <https://gurbook.com> will include all participant work materials and links to external resources. Post-it note exercises using Miro boards will be used to review approaches online. An important part of the coursework that should not be missed is finding a balance between practical online exercises and online discussions by the students. This balance makes it possible to get a full understanding of both theoretical ideas and how they can be used in the real world. Participants are also encouraged to use their own experiences when taking part in these activities. This helps them learn from their peers and gain

valuable insights. Through this approach, students learn how to effectively design for users through hands-on practice while also getting feedback from others who may have different perspectives or experience levels than themselves.

6 INSTRUCTOR BACKGROUND

Lennart E. Nacke, Ph.D., is a tenured full Professor for Human-Computer Interaction and Game Design at the University of Waterloo. He has taught courses on how to write CHI papers, gamification, and games user research at CHI and CHI PLAY for more than half a decade. He runs the successful *How to Write CHI Papers* and *The Games User Research Book* podcast and an interview series about writing CHI papers on YouTube. He runs a game UX newsletter “The Acagamic Tip Tuesday” on his website. He has served on the SIGCHI program and steering committees and taught University graduate classes on HCI research methods. Dr. Nacke has co-organized many workshops for CHI over the past five years; he also chaired the CHI PLAY 2014 & 2019 and Gamification 2013 conferences, served as the technical program co-chair for CHI PLAY 2015 and CHI Games and Play subcommittee co-chair for CHI 2017, INTERACT 2019 and CHI PLAY 2022 Full Papers Co-Chair and was the chair of the CHI PLAY steering committee until 2018. He is passionate about teaching UX research and design.

Pejman Mirza-Babaei, PhD, is the director of UXR Lab and an associate professor for human-computer interaction and user experience research at the University of Ontario Institute of Technology. He worked as the user research director at Execution Labs, Montreal. He has been involved with the GUR community since 2009, publishing more than 50 articles and co-organizing workshops and courses at international conferences. He has contributed to more than 20 published commercial games, including award-winning titles such as *pewdiepie: Legend of the Brofist*, *Crysis 2*, and *Weirdwood Manor*.

Anders Drachen, PhD, is professor and director of the Metaverse Lab, University of Southern Denmark, and a veteran data scientist. His multiple award-winning work in the game industry as well as in data science is focused on game analytics, behavioral analytics, business intelligence, game data mining, user experience, industry economics, business development, esports and Games User Research. His research and professional work is carried out in collaboration with companies across the industry.

7 RESOURCES

Participants can use the following resources to learn about the principles of user experience design and research as they apply to games.

- Games User Research Book Podcast
- Games User Research Book

8 ACCESSIBILITY

All materials will be available as recordings and written transcripts so that everyone can take part. This will enable those with hearing

impairments to access the content without difficulty. The instructors will also take extra steps to make sure that website materials are accessible for all users by providing meaningful ALT text for images and using colourblind-safe schemes. The subtitles will make it easy for everyone to follow along with online videos that were made during lessons. These measures taken together help guarantee a more inclusive course experience.

ACKNOWLEDGMENTS

Lennart Nacke’s research work was made possible by his NSERC Discovery Grant, the Canada Foundation for Innovation John R. Evans Leaders Funds, the NSERC CREATE SWaGUR grant, SSHRC Insight, Mitacs Accelerate, and CIHR.

REFERENCES

- [1] Anders Drachen, Pejman Mirza-Babaei, and Lennart Nacke. 2018. *Games User Research*. Oxford University Press, London, UK.
- [2] Pejman Mirza-Babaei, Lennart Nacke, and Anders Drachen. 2018. Games User Research Methods. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts* (Melbourne, VIC, Australia) (*CHI PLAY '18 Extended Abstracts*). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3270316.3271548>
- [3] Lennart Nacke, Anders Drachen, Kai Kuikkaniemi, Joerg Niesenhaus, Hannu J. Korhonen, Wouter M. van den Hoogen, Karolien Poels, Wijnand A. Ijsselstein, and Yvonne A. W. de Kort. 2009. Playability and Player Experience Research. In *Breaking New Ground: Innovation in Games, Play, Practice and Theory. Proceedings of DiGRA 2009*. DiGRA, London, 1–5.
- [4] Lennart E. Nacke. 2017. Games User Research and Gamification in Human-Computer Interaction. *XRDS* 24, 1 (sep 2017), 48–51. <https://doi.org/10.1145/3123748>
- [5] Lennart E. Nacke, Steve Engels, and Pejman Mirza-Babaei. 2015. Actionable Inexpensive Games User Research. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul, Republic of Korea) (*CHI EA '15*). Association for Computing Machinery, New York, NY, USA, 2461–2462. <https://doi.org/10.1145/2702613.2706681>
- [6] Lennart E. Nacke, Pejman Mirza-Babaei, and Anders Drachen. 2019. User Experience (UX) Research in Games. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (*CHI EA '19*). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3290607.3298826>
- [7] Lennart E. Nacke, Pejman Mirza-Babaei, Katta Spiel, Zachary O. Touns, and Katherine Isbister. 2018. Games and Play SIG: Engaging Small Developer Communities. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI EA '18*). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3170427.3185360>
- [8] Lennart E. Nacke, Christiane Moser, Anders Drachen, Pejman Mirza-Babaei, Andrea Abney, and Zhu (Cole) Zhenyu. 2016. Lightweight Games User Research for Indies and Non-Profit Organizations. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (*CHI EA '16*). Association for Computing Machinery, New York, NY, USA, 3597–3603. <https://doi.org/10.1145/2851581.2856504>
- [9] Magy Seif El-Nasr, Anders Drachen, and Alessandro Canossa. 2013. *Game Analytics: Maximizing the Value of Player Data*. Springer, USA.
- [10] Samantha Stahlke and Pejman Mirza-Babaei. 2022. *The Game Designer’s Playbook: An Introduction to Game Interaction Design*. Oxford University Press, London, UK.
- [11] Gareth R. White, Joonhwan Lee, Daniel Johnson, Peta Wyeth, and Pejman Mirza-Babaei. 2015. Crossing Domains: Diverse Perspectives on Players. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul, Republic of Korea) (*CHI EA '15*). Association for Computing Machinery, New York, NY, USA, 2349–2352. <https://doi.org/10.1145/2702613.2702636>